

Energy bookkeeping IDENTITY residual (not a conservation test)

Escaped trajectories

1
0.8
0.6
0.4
0.2
0

-0.1

-0.05

0

0.05

0.1

$\times 10^{-6}$

$\Delta(E_N + E_{\text{rad}} + E_S)/E_{\text{scale}}$

Entries: N = 1

Fit: degenerate Gaussian ($\sigma=0$)

$\mu = 9.6933\text{e-}08$

$\sigma = 0$

IDENTITY by construction: E_{bound} is defined

as the residual that closes this sum;

this measures roundoff + endpoint interpolation.

Independent physical residual:

$|\delta E_{\text{LL-vs-flux}}|/E_{\text{rad}}$ median/p95 = 0.48503 / 0.48503